BODYBANK

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Comprehensive Health Report

Blood Analysis + Nutrition Intelligence Report

PATIENT NAME	DATE OF REPORT	REPORT TYPE
Rahul Sharma	**13 July 2026**	**AI Health Analysis**
AGE / GENDER	**FITNESS GOAL**	**PREPARED BY**
29 / Male	Muscle Gain & Improved Energy	BodyBank AI + Medical Review

OVERALL HEALTH STATUS **Fair**

Multiple micronutrient deficiencies detected. Pre-diabetic indicators present. Immediate dietary intervention recommended.

Contents

1. Blood Test Analysis

Complete Blood Count (CBC)

MARKER	VALUE	REFERENCE RANGE	STATUS
Haemoglobin	**11.2 g/dL**	13.5–17.5 g/dL	▼ Low
RBC Count	**4.1 million/mcL**	4.5–5.5	▼ Low
WBC Count	**7,200 /mcL**	4,500–11,000	Normal
Platelets	**245,000 /mcL**	150,000–400,000	Normal
MCV	**72 fL**	80–100 fL	▼ Low

Metabolic Panel

MARKER	VALUE	REFERENCE RANGE	STATUS
Fasting Blood Glucose	**105 mg/dL**	70–99 mg/dL	▲ Elevated
HbA1c	**5.9 %**	< 5.7%	▲ Elevated
LDL Cholesterol	**128 mg/dL**	< 100 mg/dL	▲ Elevated
HDL Cholesterol	**42 mg/dL**	> 40 mg/dL	Normal
Triglycerides	**165 mg/dL**	< 150 mg/dL	▲ Elevated
Creatinine	**0.9 mg/dL**	0.7–1.2 mg/dL	Normal

Vitamins & Minerals

MARKER	VALUE	REFERENCE RANGE	STATUS
Vitamin D (25-OH)	**14 ng/mL**	30–100 ng/mL	■ Deficient
Vitamin B12	**210 pg/mL**	200–900 pg/mL	▼ Low
Ferritin	**8 ng/mL**	12–300 ng/mL	▼ Low
Zinc	**68 mcg/dL**	70–120 mcg/dL	▼ Low

Thyroid & Hormones

MARKER	VALUE	REFERENCE RANGE	STATUS
TSH	**2.1 mIU/L**	0.4–4.0 mIU/L	Normal
Testosterone (Total)	**380 ng/dL**	300–1000 ng/dL	Normal
Cortisol (AM)	**22 mcg/dL**	6–23 mcg/dL	Normal

2. Nutrition Intelligence

AVG DAILY CALORIES	AVG PROTEIN	AVG CARBS	AVG FAT
1820	68g	245g	58g
kcal/day	/day	/day	/day

MEAL QUALITY SCORE	5.8/10	Moderate — protein below target, high refined carbs observed

Most Frequently Consumed Meals

MEAL	FREQ	AVG CAL	ASSESSMENT
White Rice + Dal	6x/wk	480 kcal	Fair
Chicken Biryani	3x/wk	620 kcal	Good
Bread + Butter	5x/wk	280 kcal	Poor
Banana Milkshake	4x/wk	340 kcal	Fair

3. Clinical Interpretation

Rahul presents with iron-deficiency anaemia (Hb 11.2, Ferritin 8, MCV 72). Fasting glucose 105 + HbA1c 5.9% confirms pre-diabetic range. Vitamin D is severely deficient at 14 ng/mL, directly impacting muscle synthesis, testosterone, and immunity. Zinc deficiency compounds hormonal concerns. Cross-referencing nutrition data: protein at 68g/day is severely below the 140–160g required for muscle gain. The diet is dominated by high-glycaemic refined carbohydrates explaining borderline glucose and triglycerides.

Key Findings

! **Iron Deficiency Anaemia** Hb 11.2, Ferritin 8, MCV 72 — confirmed anaemia. Fatigue, reduced VO2max, poor recovery.

! **Severe Vitamin D Deficiency** 14 ng/mL vs target 50–80. Impacts testosterone, muscle synthesis, immunity.

! **Pre-Diabetic Indicators** Fasting glucose 105 + HbA1c 5.9% — immediate carbohydrate quality intervention required.

! **LDL & Triglycerides Elevated** LDL 128, TG 165 — early cardiovascular risk. Manageable with dietary changes.

✓ **Kidney & Thyroid Normal** Creatinine, eGFR, TSH and Free T4 all within healthy range.

4. Risk Assessment

RISK AREA	LEVEL	CONTRIBUTING FACTORS
Type 2 Diabetes	**Medium**	Fasting glucose 105, HbA1c 5.9%, high refined carbs, low fibre
Cardiovascular Disease	**Medium**	LDL 128, Triglycerides 165, low HDL relative to LDL
Muscle Gain / Performance	**High**	Protein 68g/day, iron anaemia, Vitamin D & Zinc deficiency
Chronic Fatigue	**High**	Iron deficiency, Vitamin D deficiency, B12 borderline low

5. Foods to Include

These foods directly address your confirmed deficiencies: iron anaemia, Vitamin D, pre-diabetic glucose control, and low protein. Consume these daily.

- ✓ **Chicken Liver** — Highest bioavailable iron + B12 + Zinc — 100g provides ~14mg iron
- ✓ **Whole Eggs** — Complete protein, Vitamin D (yolk), B12, and zinc in one food
- ✓ **Spinach + Citrus** — Non-haem iron absorption increases 3x when paired with Vitamin C
- ✓ **Salmon / Mackerel** — Vitamin D3, Omega-3 reduces triglycerides, high protein
- ✓ **Masoor / Moong Dal** — Iron, protein, fibre — slows glucose absorption
- ✓ **Sweet Potato** — Low GI carb with fibre — better glucose response than white rice
- ✓ **Pumpkin Seeds** — Best plant zinc source — 30g provides 30% daily zinc
- ✓ **Greek Yoghurt** — High protein, probiotics improve iron absorption

6. Foods to Avoid

The following foods directly worsen your current blood markers and should be minimised during the correction phase.

- ✗ **White Rice (large portions)** — High GI — spikes glucose. Switch to small portions + brown rice or millets.
- ✗ **White Bread / Maida products** — Refined carbs elevate triglycerides and worsen pre-diabetic pattern.
- ✗ **Tea / Coffee with meals** — Tannins reduce iron absorption by up to 70%. Avoid 1 hour around iron-rich meals.
- ✗ **Sweetened beverages** — Fructose directly raises triglycerides. Replace with water + lime.
- ✗ **Fried snacks** — Trans fats raise LDL and increase cardiovascular risk.

7. Personalised Weekly Meal Plan

Calibrated to correct iron deficiency, normalise blood glucose, and achieve protein intake for muscle gain. Target: 2,200–2,400 kcal/day with 140–160g protein.

Monday	~2240 kcal · 148g protein	
BREAKFAST	3 whole eggs scrambled + 2 multigrain toast + 1 orange	420 kcal
MID-MORNING	30g pumpkin seeds + Greek yoghurt 150g	220 kcal
LUNCH	Brown rice 1 cup + chicken curry 200g + spinach sabzi	680 kcal
SNACK	Handful almonds + 1 banana	240 kcal
DINNER	Grilled salmon 150g + sweet potato + salad	520 kcal

Tuesday	~2180 kcal · 142g protein	
BREAKFAST	Moong dal chilla 3 pieces + mint chutney + lime water	380 kcal
LUNCH	Rajma 1 cup + 2 whole wheat rotis + curd	620 kcal
SNACK	Boiled eggs x2 + walnuts 4 halves	230 kcal
DINNER	Chicken liver stir-fry 100g + broccoli + 1 roti	520 kcal

Wednesday	~2260 kcal · 151g protein	
BREAKFAST	Oats porridge 80g + whey protein 1 scoop + berries	430 kcal
LUNCH	Grilled chicken breast 200g + millet khichdi + salad	650 kcal
SNACK	Sweet potato chaat + lime juice	220 kcal
DINNER	Egg bhurji 3 eggs + 2 whole wheat rotis + dal	610 kcal

Thursday	~2200 kcal · 145g protein	
BREAKFAST	Idli 3 pieces + sambar + coconut chutney	360 kcal
LUNCH	Fish curry mackerel 150g + brown rice 1 cup + raita	660 kcal
SNACK	Pumpkin seed trail mix 40g	220 kcal
DINNER	Tofu stir-fry 200g + quinoa 80g + steamed greens	560 kcal

Friday

~2300 kcal · 158g protein

BREAKFAST	4 egg omelette + whole wheat toast + avocado	480 kcal
LUNCH	Chicken biryani brown rice 300g + mixed salad + curd	680 kcal
SNACK	Roasted chickpeas 50g + green tea	190 kcal
DINNER	Lamb keema 150g + 2 rotis + dahi	580 kcal

Saturday

~2150 kcal · 138g protein

BREAKFAST	Smoothie: spinach + banana + protein powder + almond milk	380 kcal
LUNCH	Dal makhani + 2 rotis + grilled chicken 100g + salad	640 kcal
DINNER	Prawn masala 200g + vegetable pulao + salad	540 kcal

Sunday

~2100 kcal · 135g protein

BREAKFAST	Poha with peanuts + boiled eggs x2 + lime water	420 kcal
LUNCH	Multigrain paratha 2 + curd + boiled eggs 2	550 kcal
DINNER	Chicken soup homemade + 1 roti + salad	480 kcal

8. Supplement Protocol

Based on confirmed deficiencies, the following supplements are clinically indicated. Begin under medical supervision.

SUPPLEMENT	DOSE	TIMING	REASON
Iron (Ferrous Sulphate)	60mg elemental	Morning, empty stomach + Vit C	Iron deficiency anaemia. Ferritin 8 ng/mL
Vitamin D3	60,000 IU/wk x 8wks, then 2,000 IU/day	With fatty meal	Severe deficiency 14 ng/mL
Vitamin B12 (Methylcobalamin)	500–1000 mcg	Morning with food	Borderline B12, support energy and RBC
Zinc (Zinc Picolinate)	25–30mg	With evening meal	Zinc deficiency — testosterone, protein synthesis
Omega-3 (Fish Oil)	2–3g EPA+DHA	With lunch or dinner	Reduce triglycerides 165, cardiac protection
Whey Protein	25–30g	Post-workout or morning	Bridge protein gap 68g actual vs 145g target

9. Lifestyle Recommendations

Sleep	Target 7.5–8.5 hours. Iron deficiency impairs sleep quality. Avoid screens 1 hour before bed. Magnesium glycinate 200mg before sleep improves muscle recovery.
Stress Management	Cortisol near upper limit. Chronic stress worsens glucose and impairs iron absorption. 10–15 min daily breathwork or meditation recommended.
Exercise	Moderate resistance training 4x/week. Avoid high-intensity until Hb reaches 12.5 g/dL — anaemia limits oxygen delivery. Prioritise compound lifts: squats, rows, deadlifts.
Hydration	Minimum 3–3.5 litres/day. Supports blood viscosity, kidney filtration, and reduces false glucose elevation.
Recovery	Post-workout: consume iron + protein meal within 60 minutes. Vitamin C must accompany iron meals. Avoid tea/coffee within 1 hour of meals.

10. Progress Tracking & Retest Schedule

A structured retest schedule confirms response to supplementation and dietary changes. Do not delay these — early correction prevents chronic disease.

TEST	RETEST IN	REASON
CBC — Haemoglobin & Ferritin	**8 weeks**	Monitor iron supplementation. Target Hb > 13 g/dL
Vitamin D (25-OH)	**3 months**	After high-dose weekly supplementation x8 weeks
Fasting Glucose + HbA1c	**3 months**	Confirm dietary changes reducing pre-diabetic markers
Lipid Panel	**3 months**	Omega-3 + diet should reduce TG below 150
Zinc + B12	**3 months**	Confirm therapeutic supplementation levels
Full Blood Panel	**6 months**	Comprehensive reassessment of all markers

Medical Disclaimer: *This report is generated by BodyBank.fit's AI Health System. It does not constitute a medical diagnosis. All supplement and dietary recommendations must be reviewed with your physician before implementation.*